Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

9/1/2026

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Session	**2024-25 (EVEN)**	**Course Name**	**Mobile and Web Application for IOT Lab**
Semester	**6**	**Course Code**	**23IOT1624**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**01**
Course Outcome	1. Understand and apply basic Kotlin programming constructs including variables, control structures, functions, null safety, and object-oriented concepts for Android app development. 2. Design and develop responsive user interfaces using XML layout files, ViewGroups, and widgets in Android Studio, adhering to Material Design principles. 3. Implement event handling and application logic using Kotlin to capture user input, manage app state, navigate between screens, and interact with Android system components like Toasts and Dialogs 4. Integrate learned concepts to develop a mini Android application that demonstrates use of ViewModel, multi-screen navigation, and proper UI/UX design for real-world problem-solving
Aim	Objective: Set up Kotlin development using Kotlin Playground and Android Studio
Problem Definition	Write simple Kotlin programs using variables, data types, operators. Practice control flow using if, when, for, and while.
Theory (100 words)	Kotlin is a modern, statically typed programming language developed by JetBrains and officially supported by Google for Android application development. It is designed to be concise, safe, and expressive, helping developers write clean and efficient code. Kotlin is fully interoperable with Java, allowing both languages to be used together within the same project. Variables in Kotlin are declared using val and var. The val keyword is used for immutable variables whose values cannot be changed after initialization, while var is used for mutable variables that can be modified during program execution. Kotlin supports various data types such as Int for integers, Double for decimal numbers, String for text, and Boolean for logical values. Operators in Kotlin are used to perform arithmetic and relational operations. Control flow statements such as if, when, for, and while control decision-making and repetition, enabling structured and logical program execution.

Procedure and Execution (100 Words)	Step for Implementation: ➢ Create a new Kotlin file or Android Activity. ➢ Declare variables using val (immutable) and var (mutable). ➢ Use basic data types like Int, Double, String, and Boolean. ➢ Apply operators such as arithmetic (+) and relational (>=). ➢ Use if to check conditions. ➢ Use when for multiple condition checking. ➢ Use for loop to iterate a range. ➢ Use while loop to repeat statements based on a condition. ➢ Display the output using println() or TextView.
	Code: **Kotlin Playground**

```kotlin
fun main() {
    // Variables
    val name: String = "Sheikh Wasimuddin"   // Immutable
    var age: Int = 21            // Mutable

    // Data Types
    val percentage: Double = 87.5
    val isPassed: Boolean = true

    // Operators
    val nextAge = age + 1
    val marks = 450
    val total = 500
    val result = marks >= 200

    println("Name: $name")
    println("Age: $age")
    println("Next Age: $nextAge")
    println("Percentage: $percentage")
    println("Passed: $isPassed")
    println("Result: $result")
}
```

```kotlin
fun main() {
    val number = 7

    // if-else
    if (number % 2 == 0) {
        println("Even number")
    } else {
        println("Odd number")
    }

    // when statement
    when (number) {
        1 -> println("One")
        2 -> println("Two")
        in 3..10 -> println("Between 3 and 10")
        else -> println("Other number")
    }

    // for loop
    for (i in 1..5) {
        print("$i ")
    }
    println()

    // while loop
    var count = 5
    while (count > 0) {
        print("$count ")
        count--
    }
}
```

```kotlin
package com.example.kotlinbasics

import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        val textView = TextView(this)
        setContentView(textView)

        // Variables and Data Types
        val name = "SHEIKH WASIMUDDIN"
        var age = 21
        val pi = 3.14
        val isStudent = true

        // Operators
        val sum = age + 5
        val isAdult = age >= 18
```

```
        // Control Flow
        val result = if (isAdult) "Adult" else "Minor"

        val day = 3
        val dayName = when (day) {
            1 -> "Monday"
            2 -> "Tuesday"
            3 -> "Wednesday"
            else -> "Other day"
        }

        var loopResult = "For Loop: "
        for (i in 1..5) {
            loopResult += "$i "
        }

        var whileResult = "While Loop: "
        var count = 3
        while (count > 0) {
            whileResult += "$count "
            count--
        }

        textView.text = """
            Name: $name
            Age: $age
            Pi: $pi
            Student: $isStudent
            Sum: $sum
            Status: $result
            Day: $dayName

            $loopResult
            $whileResult
        """.trimIndent()
    }
}
```

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Output:

Kotlin Playground

```kotlin
fun main() {
    // Variables
    val name: String = "Sheikh Wasimuddin"   // Immutable
    var age: Int = 21            // Mutable

    // Data Types
    val percentage: Double = 87.5
    val isPassed: Boolean = true

    // Operators
    val nextAge = age + 1
    val marks = 450
    val total = 500
    val result = marks >= 200

    println("Name: $name")
    println("Age: $age")
```

```
Name: Sheikh Wasimuddin
Age: 21
Next Age: 22
Percentage: 87.5
Passed: true
Result: true
```

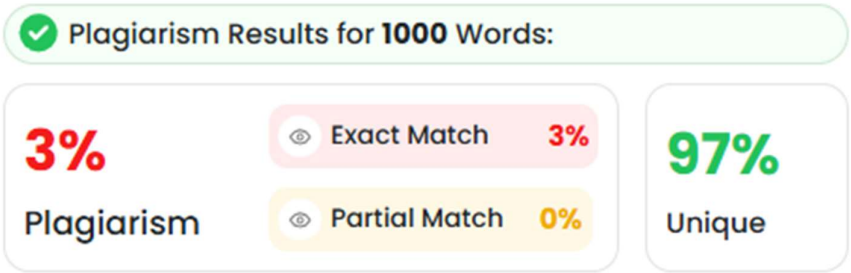
```kotlin
fun main() {
    val number = 7

    // if-else
    if (number % 2 == 0) {
        println("Even number")
    } else {
        println("Odd number")
    }

    // when statement
    when (number) {
        1 -> println("One")
```

```
Odd number
Between 3 and 10
1 2 3 4 5
5 4 3 2 1
```

Output Analysis	The program successfully demonstrates the use of various data types, conditional statements, and looping constructs. The correctness of the program execution is verified through the following outputs: 1. **Name: Sheikh Wasimuddin** This output displays the value stored in the String variable name. It confirms that the variable is correctly initialized and printed. 2. **Age: 21** This line shows the value of the integer variable age, representing the current age. It verifies correct usage of the Integer data type. 3. **Pi: 3.14** This output represents a Double data type value, confirming that the program handles decimal values accurately. 4. **Student: true** This line displays the Boolean variable value, indicating that the student status is set to true, validating Boolean data handling. 5. **Sum: 26** This output is the result of an arithmetic operation where 5 is added to the age (21 + 5). It confirms correct implementation of arithmetic operators. 6. **Status: Adult** This result is obtained using an if conditional statement. Since the age is greater than or equal to 18, the program correctly classifies the person as an adult. 7. **Day: Wednesday** This output is generated using the when statement. The numeric value 3 correctly maps to "Wednesday," demonstrating proper multiple-condition checking. 8. **For Loop: 1 2 3 4 5** This line is produced by the for loop, which iterates from 1 to 5 and prints each value in sequence, confirming correct loop execution. 9. **While Loop: 3 2 1** This output is produced by the while loop, which continues

	execution while the condition is true and decrements the value each time, demonstrating proper loop control.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	The experiment successfully demonstrates the fundamental concepts of programming using variables, data types, conditional statements, and looping constructs. By executing the program, correct initialization and usage of String, Integer, Double, and Boolean data types were verified. The use of if and when statements enabled effective decision-making based on given conditions, while the for and while loops confirmed proper iteration and control flow. Overall, the program executed without errors and produced the expected output, reinforcing a clear understanding of basic programming logic and control structures essential for problem-solving and application development.
Plag Report (Similarity index < 12%)	
Date	**09/01/2026**